

Ammonia 4-Stroke Gas Engine Study

Joint YTL x MYSGH2 timeline — maturity, carbon emissions & performance-contract assessment across the shortlisted OEM set | Kickoff: **Tue 25 Aug 2026** → Finish: **Tue 22 Sep 2026** (4 calendar weeks / 20 working days)

This document details the work timeline agreed between YTL (IPP, Singapore) and MYSGH2 to study the maturity, carbon-emission performance, and contractual (performance-guarantee) structure of 4-stroke ammonia gas engines across the shortlisted OEM set. It expands the milestone plan into day-level tasks with owners and dependencies, mirroring the accompanying Gantt-chart workbook.

1. Milestone Summary

Date	Milestone	Description
Fri 21 Aug 2026	Prep complete	OEM shortlist, kickoff charter, RFI template, and NDA status all confirmed ahead of kickoff.
Tue 25 Aug 2026	KICKOFF MEETING	Formal study kickoff with YTL: objectives, scope, timeline and success criteria aligned.
Fri 28 Aug 2026	RFI issued to all OEMs	Data-request package (maturity evidence, emissions data, commercial terms) sent to every shortlisted OEM.
Fri 4 Sep 2026	Week 2 status review	Progress check-in with YTL: OEM Round-1 discussions and framework v1 status.
Fri 11 Sep 2026	Week 3 status review	Progress check-in with YTL: maturity scorecard and carbon comparison populated.
Mon 21 Sep 2026	Draft report issued	Consolidated draft report (maturity, carbon, performance-contract framework, risk register) submitted for YTL review.
Tue 22 Sep 2026	FINAL REPORT & CLOSE-OUT	YTL comments incorporated; final report delivered; close-out session presenting findings and recommended next steps.

2. Detailed Task Timeline

Full task breakdown by phase, with owner, working-day duration and predecessor(s). Milestones are shaded gold; task rows are shaded by workstream (see legend, §4).

Phase	ID	Task	Owner	Start	Finish	Dur. (wd)	Predecessor(s)
Prep	P1	Finalize OEM shortlist & contacts	MYSGH2	Mon 17-Aug	Tue 18-Aug	2	-
Prep	P2	Draft kickoff agenda & study charter	MYSGH2	Mon 17-Aug	Wed 19-Aug	3	-
Prep	P3	Draft RFI / data-request template	MYSGH2	Tue 18-Aug	Thu 20-Aug	3	P1
Prep	P4	Confirm NDA status per OEM	YTL	Wed 19-Aug	Fri 21-Aug	3	P1
Prep	P5	Internal YTL/MYSGH2 alignment call	Both	Fri 21-Aug	Fri 21-Aug	1	P2, P3, P4
Week 1	M1	KICKOFF MEETING	Both	Tue 25-Aug	Tue 25-Aug	1	P5

Phase	ID	Task	Owner	Start	Finish	Dur. (wd)	Predecessor(s)
Week 1	W1.1	Agree assessment framework skeleton	MYSGH2	Wed 26-Aug	Thu 27-Aug	2	M1
Week 1	W1.2	Issue RFI to all OEMs	MYSGH2	Fri 28-Aug	Fri 28-Aug	1	P3, M1
Week 1	W1.3	Finalize workstream owners & roles	Both	Mon 31-Aug	Mon 31-Aug	1	M1
Week 2	W2.1	OEM 1:1 discussions - Round 1	MYSGH2	Tue 01-Sep	Thu 03-Sep	3	W1.2
Week 2	W2.2	Build maturity assessment framework v1	MYSGH2	Tue 01-Sep	Fri 04-Sep	4	W1.1
Week 2	W2.3	Build carbon-emission methodology v1	MYSGH2	Wed 02-Sep	Mon 07-Sep	4	W1.1
Week 2	W2.4	Weekly status check-in (Wk2)	Both	Fri 04-Sep	Fri 04-Sep	1	W2.1
Week 3	W3.1	OEM discussions - Round 2 (deep-dive)	MYSGH2	Tue 08-Sep	Thu 10-Sep	3	W2.1
Week 3	W3.2	Populate per-OEM maturity scorecard	MYSGH2	Wed 09-Sep	Fri 11-Sep	3	W2.2, W3.1
Week 3	W3.3	Populate carbon emission comparison table	MYSGH2	Wed 09-Sep	Fri 11-Sep	3	W2.3, W3.1
Week 3	W3.4	Draft performance-contract framework	MYSGH2/YTL	Thu 10-Sep	Mon 14-Sep	3	W3.1
Week 3	W3.5	Weekly status check-in (Wk3)	Both	Fri 11-Sep	Fri 11-Sep	1	W3.2, W3.3
Week 4	W4.1	Build risk & derisking framework	MYSGH2	Tue 15-Sep	Thu 17-Sep	3	W3.2, W3.3, W3.4
Week 4	W4.2	OEM follow-ups / gap-fill queries	MYSGH2	Tue 15-Sep	Wed 16-Sep	2	W3.1
Week 4	W4.3	Consolidate all findings	MYSGH2	Thu 17-Sep	Fri 18-Sep	2	W4.1, W4.2
Week 4	W4.4	Draft consolidated report	MYSGH2	Fri 18-Sep	Mon 21-Sep	2	W4.3
Week 4	W4.5	Internal YTL review of draft report	YTL	Mon 21-Sep	Mon 21-Sep	1	W4.4
Finish	F1	Incorporate YTL review comments	MYSGH2	Tue 22-Sep	Tue 22-Sep	1	W4.5
Finish	F2	FINAL REPORT & CLOSE-OUT SESSION	Both	Tue 22-Sep	Tue 22-Sep	1	F1

3. Workstream × Week View

Workstream	Prep (17-21 Aug)	Wk1 (25-31 Aug)	Wk2 (1-7 Sep)	Wk3 (8-14 Sep)	Wk4 (15-21 Sep)	Finish (22 Sep)
Governance	Alignment call	Kickoff	Status check-in	Status check-in	Internal review	Close-out
OEM Engagement	RFI drafted, NDAs confirmed	RFI issued	Round 1 calls	Round 2 deep-dive	Follow-ups / gap-fill	-
Maturity	-	Framework skeleton	Framework v1	Scorecard populated	Consolidated	Finalized
Carbon	-	-	Methodology v1	Comparison table	Consolidated	Finalized
Contract	-	-	-	Framework v1 (guarantees, LDs)	Aligned w/ commercial	Finalized
Risk	-	-	-	-	Register + mitigations	Finalized
Reporting	-	-	-	-	Draft report	Final report

4. Legend

	Category	Description
	Milestone	Decision point, checkpoint, or deliverable hand-off
	Governance	Kickoff, status check-ins, internal alignment & sign-off
	OEM Engagement	RFI issuance, OEM discussions, deep-dive Q&A, follow-ups
	Maturity	TRL/CRI-style maturity assessment framework & OEM scorecards
	Carbon	Carbon-emission methodology (incl. N2O/GWP273, NOx, NH3 slip) & comparison
	Contract	Performance-contract framework: availability, efficiency, emissions guarantees, LDs, warranty
	Risk	Risk & derisking framework: technology, fuel-supply, safety/toxicity, regulatory, counterparty risk
	Reporting	Findings consolidation, draft & final report production

5. Notes & Assumptions (per CLAUDE.md §7 No-Fabrication Rule)

- This is a **proposed schedule**, not a sourced technical claim — every OEM-facing date is provisional until calendar-confirmed with that OEM.
- **Assumption:** OEM 1:1 discussions can be scheduled within Week 2–3 (1–11 Sep); actual availability depends on each OEM's calendar and NDA status.

- **Assumption:** "Performance contract" is scoped as the **framework** for guarantee structure (availability, efficiency/heat-rate, emissions, warranty, liquidated damages) — not a negotiated term sheet with any specific OEM. Confirm with YTL if the latter is wanted by 22 Sep, as that widens scope.
- **Assumption:** Maturity assessment uses a TRL-style (or CRI-equivalent) scale populated from OEM-supplied evidence (test hours, reference installations, certification status), cross-checked against the sourced OEM data already held in the MYSGH2 project memory (Wartsila 25 Ammonia, MAN AmmoniaMot 2, Baker Hughes/Hanwha NovaLT16, IHI IM270, Mitsubishi Power H-25).
- Carbon methodology must explicitly include **N₂O** (GWP $\approx 273 \times \text{CO}_2$, 100-yr) and **NH₃ slip**, not just CO₂/NO_x — the ammonia-specific risk generic combustion-emissions templates typically miss.
- Four calendar weeks is **tight** for OEM-by-OEM technical deep-dives across a full shortlist. If the OEM list is long (5+), consider parallel-track OEM calls (multiple MYSGH2/YTL reviewers) in Weeks 2–3, or flag a scope/timeline trade-off to YTL at kickoff.
- Working-day durations exclude weekends; the accompanying Gantt-chart workbook (*YTL_Ammonia_Gas_Engine_Study_Gantt.xlsx*) computes duration via NETWORKDAYS() and will recalculate automatically if start/finish dates are edited.

Source: MYSGH2 project timeline, drafted 2026-08-14, anchored to the agreed kickoff (25 Aug 2026) and finish (22 Sep 2026) dates. Companion files: power_generation/YTL_ammonia_gas_engine_OEM_study_timeline.md, YTL_Ammonia_Gas_Engine_Study_Gantt.xlsx.